Lucas Bucht

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Education

University of Pittsburgh — Pittsburgh, PA
Bachelor of Science in Electrical Engineering — 2024 - Present
Expected May 2028

Technical Skills

Analog & Small-Signal Design: Op-amp circuit design (LM386), signal path analysis, frequency response & filter/tone-shaping design, noise reduction & stability testing

Circuit Simulation & Layout: LTspice (SPICE simulation), Altium (schematic capture)

Lab & Debugging: Oscilloscope & multimeter-based debugging, protoboard prototyping & soldering

Digital Logic & HDL: VHDL, datapath design (ALU, registers, decoders, multiplexers), Siemens HDL Designer, ModelSim (testbench verification)

Embedded & FPGA: Quartus Prime, FPGA synthesis (DE10-Lite), Arduino IDE (C/C++)

CAD & Fabrication: Fusion 360, 3D printing, CNC, laser cutting

Tools: Git/GitHub, technical documentation

Engineering Projects

Guitar Fuzz Pedal (Personal Project) — Spring 2026

- Designed the full analog signal path using circuit analysis for a LM386 op-amp-based pedal.
- Synthesized the amplification and frequency response in LTspice and Altium to verify component choices.
- Prototyped and soldered on a protoboard, testing across 4 iterations to reduce noise and improve stability.
- Created a scale enclosure in Fusion 360 to plan out jack placement and mounting points.

VHDL Calculator – ECE 0201 (Digital Electronics Project) — Fall 2025

- Built a complete datapath in Siemens HDL Designer, including a 4x8 register and ALU.
- Developed core components like an 8-bit register, 2:4 decoder, and multiplexer.
- Wrote 10+ do-file testbenches covering all datapath components and verified functionality in ModelSim.
- Synthesized the design to a DE10-Lite FPGA in Quartus Prime.

SSOE First Year Engineering Conference: Perovskite Solar Cells — Spring 2025

- Served as the lead writer for a 12-page analysis of the efficiency, cost, and sustainability advantages over silicon.
- Utilized engineering research databases and technical writing standards to produce a professional academic report.

UV-C Light Bar Waste Reduction System — Fall 2024

- Designed a sustainable solution to reduce single-use disposable wipes in on-campus gyms.
- Led CAD development across 5+ design iterations in Fusion 360, collaborating on a 3-person team.
- Programmed a virtual prototype using the Arduino IDE.

Experience

Engineering Makerspace Mentor — University of Pittsburgh
September 2026 - Present

- Trained students on safe operation of 3D printers, CNC machines, and laser cutters for project fabrication.
- Enforced equipment safety protocols during 4+ hours of weekly open shop coverage.
- Guided students through technical troubleshooting to keep projects on track.

Panther Leadership Academy Peer Facilitator — University of Pittsburgh
September 2026 - Present

- Selected as a peer facilitator to lead weekly leadership development sessions for program participants.
- Mentored 20 students on leadership philosophies, communication, conflict resolution, and group dynamics.

Resident Assistant — University of Pittsburgh
August 2025 - Present

- Supported 40+ residents through conflict-resolution, crisis response, and community engagement.
- Organized 10 educational and social programs per semester to promote student success and well-being.

Floor Staff – Regal Cinemas — Crofton, MD
May - August 2025

- Provided high-volume customer service and supported theater operations during peak hours.
- Mentored new hires by sharing operational knowledge and helping them acclimate to team workflows.